

Table 2: Device management, visualisation and analytics of the IoT platforms

IoT platform	Device management	Data visualisation	Analytics	Multi-tenancy
Arduino IoT Cloud	Yes	Yes	Yes	-
OpenRemote	Yes	Yes	Yes	Yes
Mainflux	Yes	Grafana	Yes	Yes
ThingsBoard	Yes	Dashboard widgets	Yes	Yes
Thingier.io	Yes	Grafana	Yes	Yes
Kaa	Yes	Direct/IoT gateway	Yes	Yes
Blynk	Yes	Yes	Yes	Yes
SiteWhere	Yes	Grafana	Can detect outliers too	Yes
ThingSpeak	Yes (with MATLAB)	MATLAB visualisation app	Yes	-
Zetta	Yes	Yes	Yes	Yes

Plans include Kaa Cloud 5, Kaa Cloud 15, and Kaa Cloud 100, of which only Kaa Cloud 5 offers a 14-day free trial on up to five connected devices with multi-user access.

Blynk

URL: <https://blynk.io/>

Blynk helps to build and manage connected hardware. It offers device provisioning, sensor data visualisation, remote control with mobile and web applications, over-the-air firmware updates, secure cloud, data analytics, user and access management, alerts and automations. The Blynk IoT platform prototypes, deploys, and remotely manages connected electronic devices at any scale.

Blynk powers low-batch manufacturers of smart home products, complex HVAC systems, agricultural equipment, and everyone in between. It offers two plans -- Free and Business.

SiteWhere

URL: <https://sitewhere.io/en/>

SiteWhere provides a multi-tenant

microservice-based infrastructure that includes the key features required to build and deploy IoT applications. SiteWhere infrastructure and microservices are deployed on Kubernetes, allowing for deployment on-premise or almost with any cloud provider. It runs on Apache Tomcat and offers MongoDB and HBase implementations. You can deploy SiteWhere to cloud platforms like AWS, Azure, GCP, or on-premise.

ThingSpeak

URL: <https://thingspeak.com/>

ThingSpeak is an IoT analytics platform service that allows you to aggregate, visualise, and analyse live data streams in the cloud. You can send data to ThingSpeak from your devices, create instant visualisation of live data, and send alerts. Data collection in the cloud

with advanced data analysis using MATLAB is possible.

Zetta

URL: <https://www.zettajs.org/>

Zetta is an open source platform built on Node.js for creating IoT servers that run across geo-distributed computers and the cloud. Zetta combines REST APIs, WebSockets and reactive programming perfect for assembling many devices into data-intensive, real-time applications. It is a server-oriented platform with a flow-based reactive programming development philosophy and connected by cloud services to develop IoT frameworks. These cloud services include visualisation tools and support for machine analytics tools like Splunk. It creates a zero-distributed network by connecting endpoints such as Linux and Arduino hacker boards with platforms such as Heroku. 

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